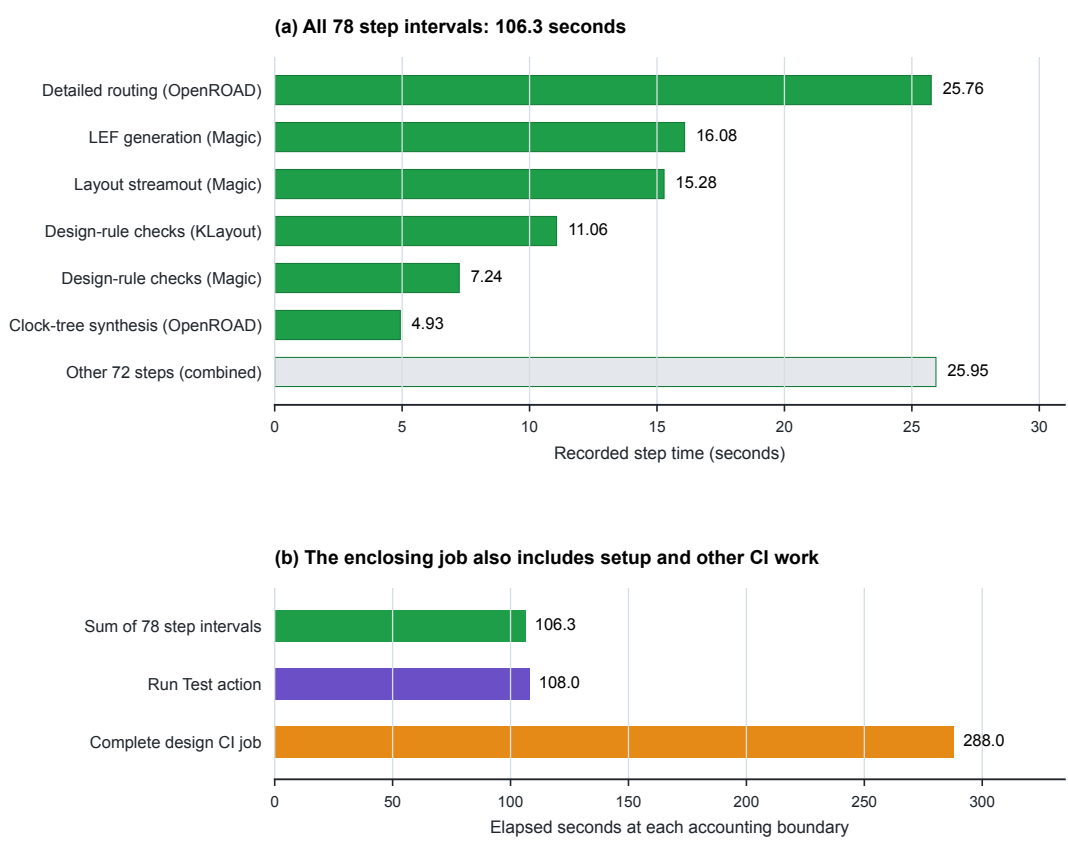
(a) All 78 step intervals: 106.3 seconds
Detailed routing (OpenROAD)
LEF generation (Magic)
Layout streamout (Magic)
Design-rule checks (KLayout)
Design-rule checks (Magic)
Clock-tree synthesis (OpenROAD)
Other 72 steps (combined)
25.76
16.08
15.28
11.06
7.24
4.93
25.95
0
5
10
15
20
25
30
Recorded step time (seconds)
(b) The enclosing job also includes setup and other CI work
Sum of 78 step intervals
Run Test action
Complete design CI job
106.3
108.0
288.0
0
50
100
150
200
250
300
Elapsed seconds at each accounting boundary